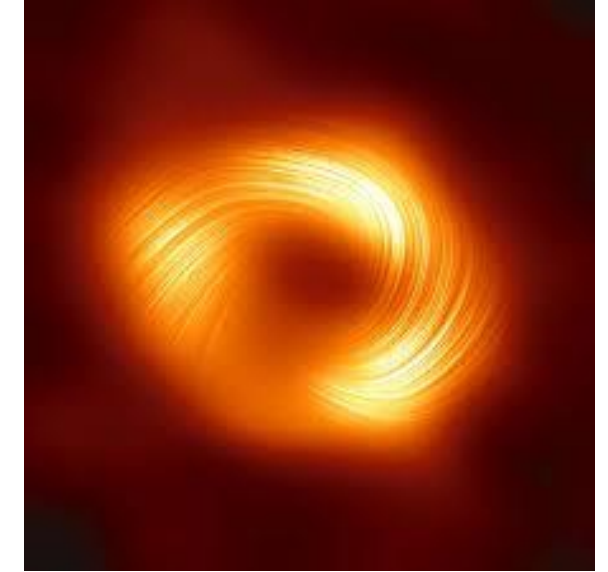
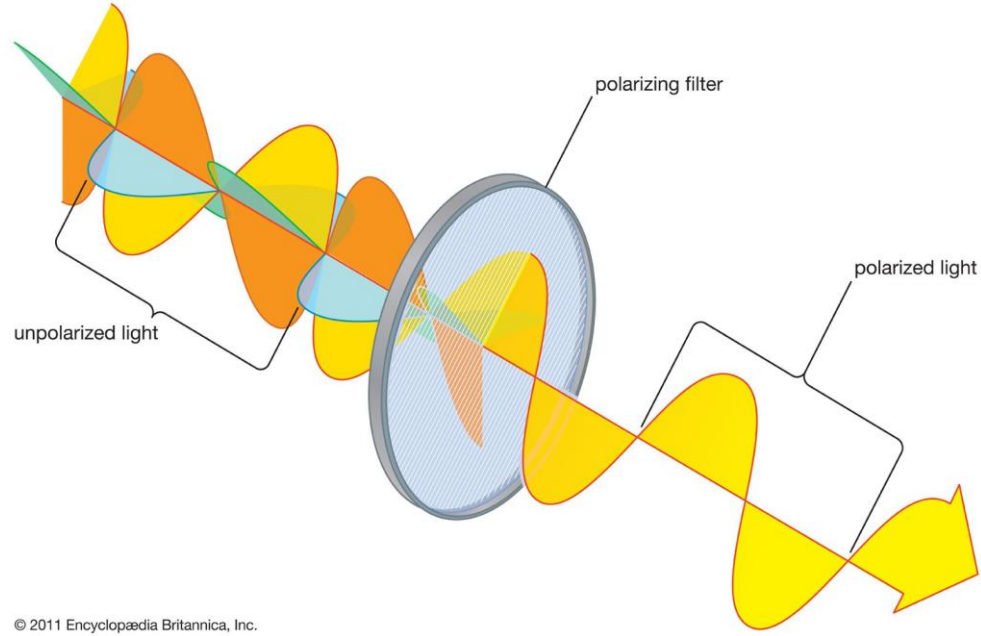


Wave and Optics

Part-7



Polarized light reveals secrets of black hole's magnetic field

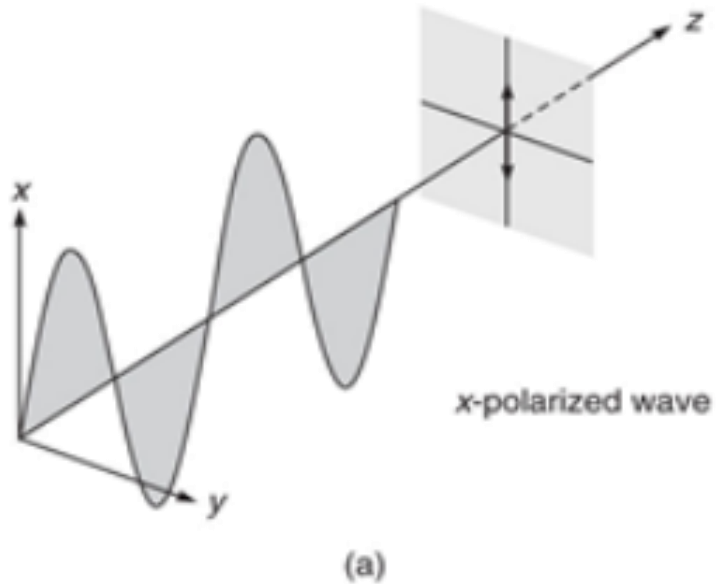
Dr. Ram Prakash
Professor
Department of Physics
Indian Institute of Technology (IIT), Jodhpur

Email: ramprakash@iitj.ac.in

Polarization is a property of transverse waves which specifies the geometrical orientation of the oscillations.

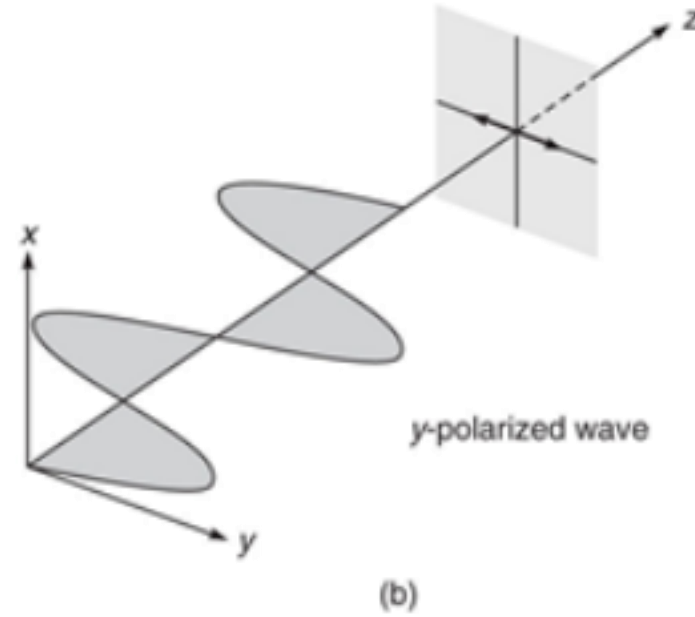
Polarization

Consider a transverse wave, generated by moving a string up & down



(a) The wave is traveling along z and is confined in the xz plane

$$\vec{E} = \hat{x} E_o \cos(kz - \omega t + \varphi_o) ; k = \frac{2\pi}{\lambda_o} n$$



(b) The wave is traveling along z and is confined in the yz plane

$$\vec{E} = \hat{y} E_o \cos(kz - \omega t + \varphi_o) ; k = \frac{2\pi}{\lambda_o} n$$

- Unpolarized wave $\Rightarrow$ $\mathbf{E}$ is vibrating randomly in all directions transverse to the z direction



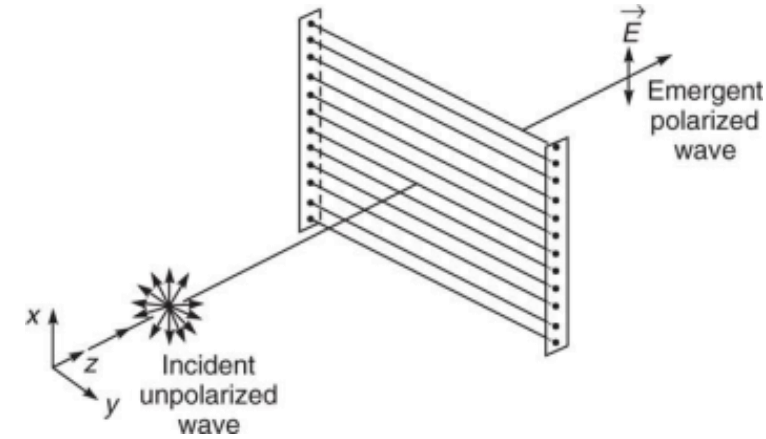
- Consider an unpolarized wave to be incident on a wire grid polarizer
- Assuming the separation between the wires, which are very thin is $\sim \lambda$, x component of $\mathbf{E}$ passes through!

$\Rightarrow$ the em wave becomes linearly polarized with $\mathbf{E}$ along $\hat{x}$

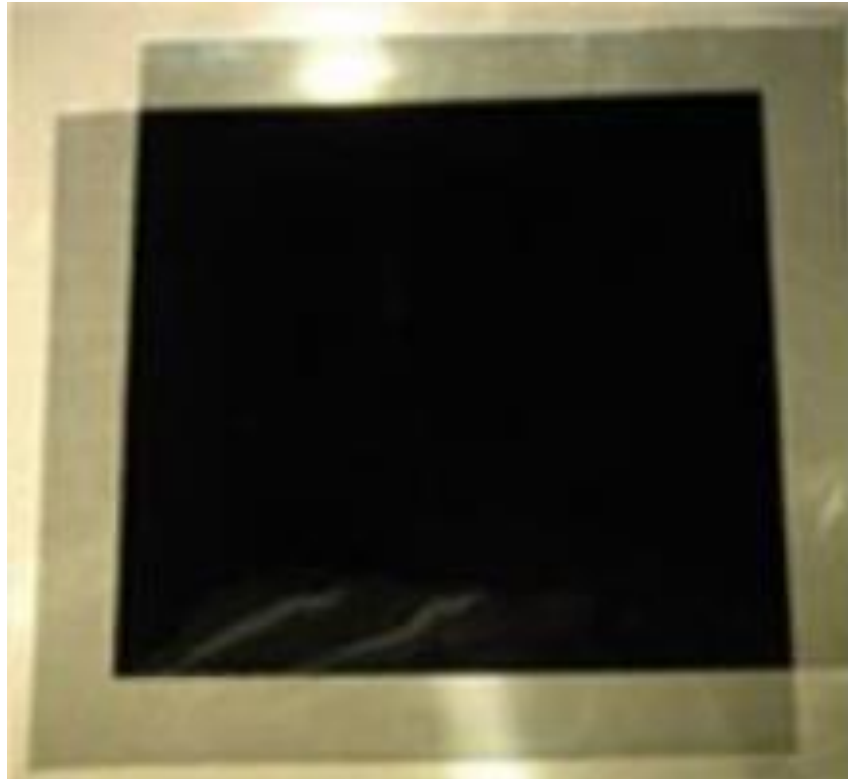
$\Rightarrow$ Component parallel to the chain of vibrating $\mathbf{E}$ along y gets absorbed due to Joule heating

- Component of vibrating $\mathbf{E}$ along y gets absorbed due to Joule /resistive heating in the conducting wire

$$P = I^2 R \quad \text{Where } P \text{ is power of heating}$$



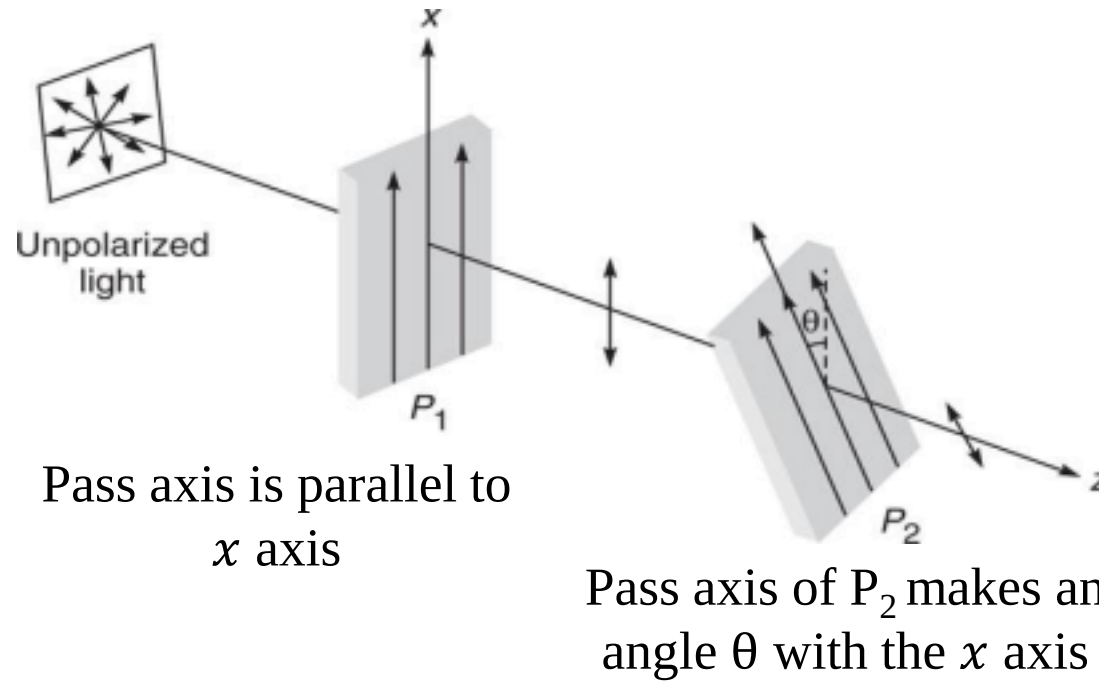
- At optical frequencies, since the λ is very short $\sim$ sub-micrometer, it is very difficult to fabricate such a wire grid polarizer
- Such a polarizer can instead be produced through use of long chain polymer molecules containing atoms of, e.g., iodine that can provide high conductivity along the length of the chain.



Called a **polaroid** sheet

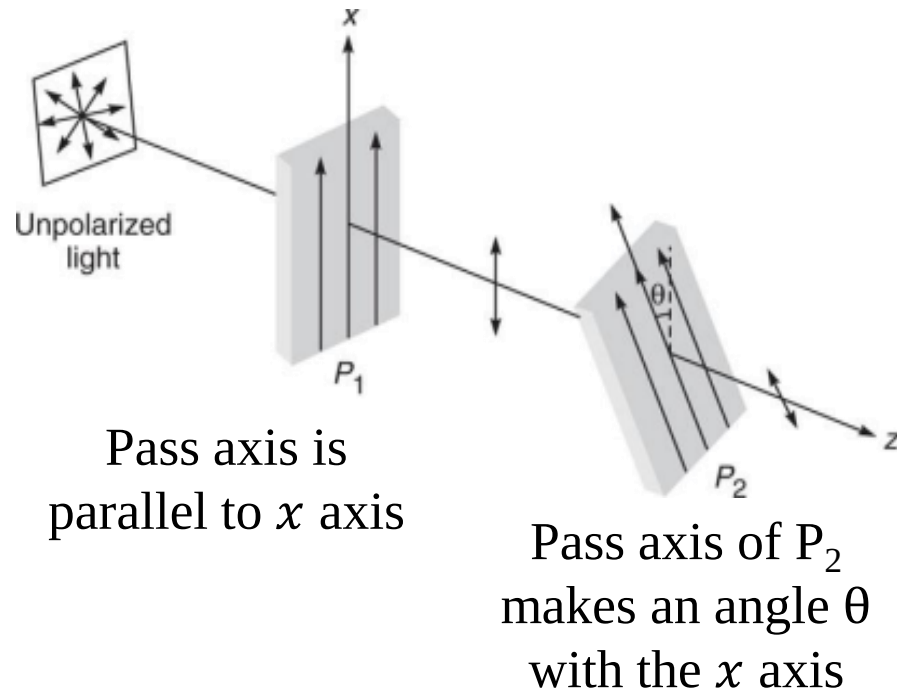
Malus's law:

$$I = I_0 \cos^2 \theta$$



⇒ Let's assume input unpolarized light, which becomes polarized, after exiting through P_1 , with its E field oscillating along the x direction is allowed to be incident on polaroid P_2

❖ If P_2 is rotated about the z axis, then how will the intensity of the emergent beam will vary?



- It will vary according to Malus law.
- If rotated clockwise, till its pass axis is $\parallel$ to x axis, its intensity will be maximum
- On further clockwise rotation, its intensity will keep on decreasing till its pass axis becomes $\parallel$ to y axis when its intensity will be almost 0
- Any further clockwise rotation will lead to increase in its intensity till it becomes max and followed by a gradual decrease in will keep on decreasing till its pass axis becomes $\parallel$ to y axis when its intensity will be almost 0 again and so on

Consider time variation of the superposition of two waves :

?

$$\vec{E} = \hat{x} a \cos(kz - \omega t) ; k = \frac{2\pi}{\lambda_o} n$$

$$\vec{E} = \hat{y} b \cos(kz - \omega t + \theta) ; k = \frac{2\pi}{\lambda_o} n$$

at an arbitrary plane $z = 0$ perpendicular to the z axis.

Superposition of the two will yield resultant field: $\vec{E} = \vec{E}_x + \vec{E}_y$

$\Rightarrow$ corresponding waves at $z = 0$:

$$E_x = a \cos(-\omega t) = a \cos \omega t$$

$$E_y = b \cos(-\omega t + \theta)$$

Consider $\theta = p\pi$; $p = 0, \pm 1, \pm 2, \dots$

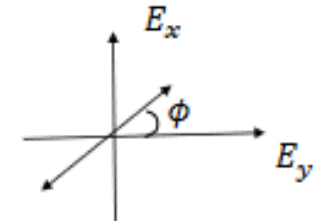
For $\theta = 0$, $E_y = b \cos(-\omega t) = b \cos(\omega t)$

$\Rightarrow \frac{E_x}{E_y} = \frac{a}{b}$ (independent of t);

$\Rightarrow E_x = \frac{a}{b} E_y \equiv m E_y \Rightarrow$ Eq. of a st line in the $E_x - E_y$ plane making an angle ϕ with the E_y axis

↓
Slope

↓
 $\tan^{-1} \left(\pm \frac{a}{b} \right)$



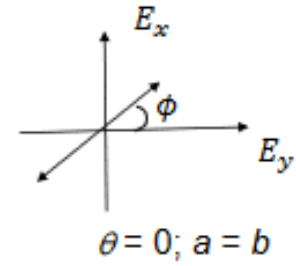
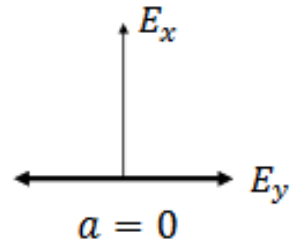
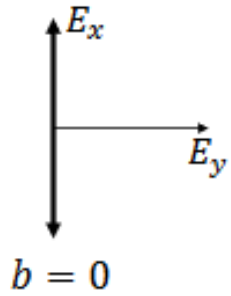
$\theta = 0$; $a = b$

Linearly polarized (LP)

If $b = 0$, $\theta = 0$

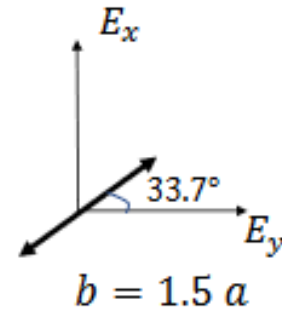
then $\phi = \pi/2$

$\Rightarrow$



If $b = 1.5a \Rightarrow \frac{a}{b} = \frac{1}{1.5} = 0.667$

$\Rightarrow \tan^{-1}(0.667) = 33.7^\circ$



Let us examine when $\theta = p\pi$; $p = 0, \pm 1, \pm 2, \dots$

Consider $p = \pm 1$:

$$\begin{aligned} E_y &= b \cos(-\omega t \oplus \pi) \Rightarrow E_y = b \cos [(-\omega t + \pi)] \\ &= b \cos(\omega t - \pi) = b [(\cos \omega t \cos \pi + \sin \omega t \sin \pi)] \\ &= -b \cos \omega t \end{aligned}$$

Likewise for $p = -1$:

$$\begin{aligned} E_y &= b \cos (-\omega t - \pi) \\ &= b \cos [-(\omega t + \pi)] = b \cos (\omega t + \pi) \\ &= -b \cos \omega t \end{aligned}$$

Since $E_x = a \cos \omega t$

Thus for $p = \pm 1$:

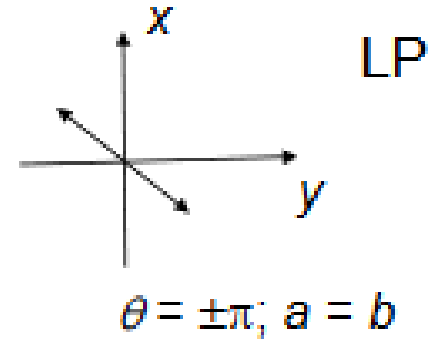
$$E_x = -\frac{a}{b} E_y$$

Likewise for $p = 0, \pm 2$ i.e. even:

$$E_x = +\frac{a}{b} E_y$$

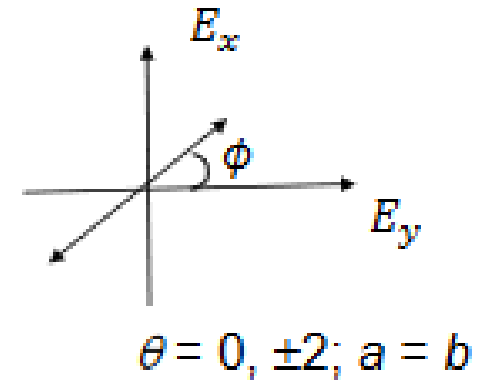
- : $p = \text{odd}$ (out of phase)

+ : $p = \pm \text{even}$ (in phase)



$\Rightarrow$ In general we can write

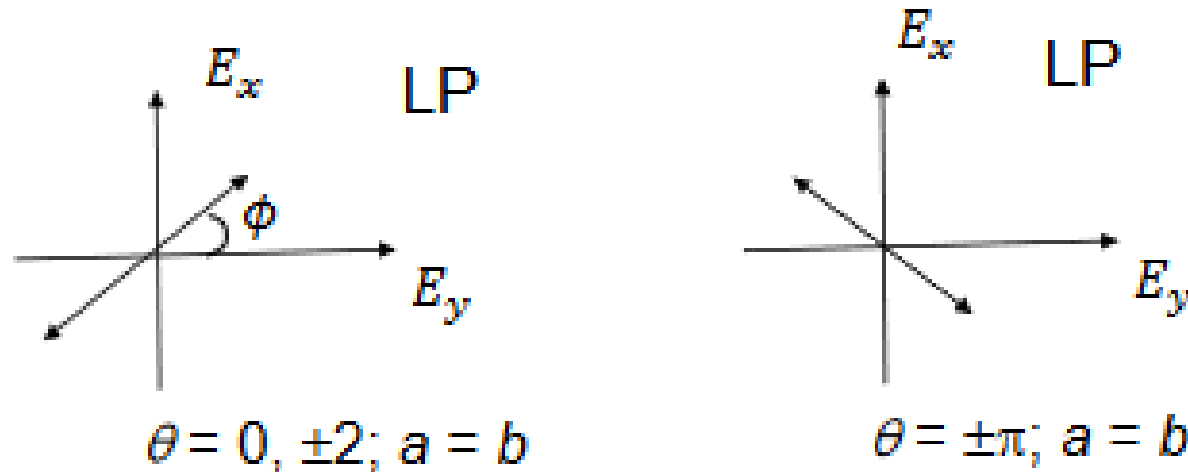
$E_y = (-1)^p b \cos \omega t$; when $p=0$, even: +1 (two waves are in phase)



& when $p = \text{odd}$: -1 (two waves are out of phase)

For $p = 0, \pm 2, \pm 4, \dots$ Two waves are in phase

For $p = \pm 1, \pm 3, \dots$ Two waves are π out of phase



$\Rightarrow$ Superposition of two linearly polarized waves in phase (or π out of phase) with their $\mathbf{E}$ at right angles to each other is again a linearly polarized wave with its $\mathbf{E}$ oscillating in a new direction.

State of Polarization

If $\theta \neq p \pi$

in general resultant $\mathbf{E}$ will not be along a straight line

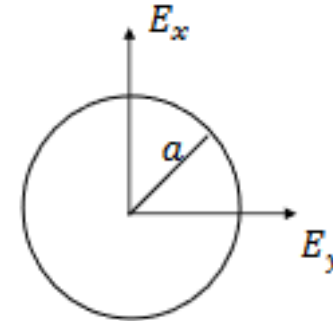
Consider $\theta = \frac{\pi}{2}$, and $a = b$

i.e. the two superposed waves are of equal amp:

$$E_x = a \cos \omega t$$

$$E_y = a \sin \omega t$$

$$\Rightarrow E_x^2 + E_y^2 = a^2$$



This is eq of a circle of radius a

Thanks